Vellore – 632 014, Tamil Nadu, India.

SCHOOL OF ADVANCED SCIENCES

DEPARTMENT OF MATHEMATICS

FALL SEMESTER – 2025~2026

BMAT205L – Discrete Mathematics and Graph Theory

QUESTION BANK

Modules 1, 2, 3 and 4

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The course coordinators thankfully acknowledge the question bank preparation committee in-charges and members for their significant contribution in bringing out of this document. The queries (if any) related to this question bank can be clarified with the respective module wise in-charges.

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Module 1
Mathematical Logic

1. Write each of these propositions in the form "p if and only if q" in English.
   a) For you to get an A in this course, it is necessary and sufficient that you learn how to solve discrete mathematics problems.
   b) If you read the newspaper every day, you will be informed and conversely.
   c) It rains if it is a weekend day, and it is a weekend day if it rains.
   d) You can see the wizard only if the wizard is not in, and the wizard is not in only if you can see him.

2. Obtain the principal disjunctive and principal conjunctive normal forms of the following statement.
   $S \Leftrightarrow (P \rightarrow (Q \wedge R)) \wedge (\neg P \rightarrow (\neg Q \wedge \neg R))$.

3. Verify the equivalence of the following.
   $(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$.

4. Test whether the premises $P \rightarrow Q$, $Q \rightarrow R$, $S \rightarrow R$, and $P \wedge S$ are consistent.

5. Obtain PCNF and PDNF of the statement formula $(P \wedge Q) \vee (\neg P \wedge Q \wedge R)$.

6. Prove Demorgan's laws in two variables and compute equivalent of the following using Demorgan's law:
   "It is not the case that roses are red and violets are blue."

7. Show the following equivalent formula without constructing the truth table:
   $(P \vee Q) \vee (Q \vee \neg R) \wedge (P \vee R) \Leftrightarrow (P \vee Q)$

8. Show that $(S \vee R)$ is tautologically implied from $P \vee Q$, $P \rightarrow R$ and $Q \rightarrow S$.

9. (*i*). Derive $p \rightarrow (q \rightarrow s)$ from the premises $p \rightarrow (q \rightarrow r)$ and $q \rightarrow (r \rightarrow s)$.
   (*ii*). Show that $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$ are not logically equivalent.

10. Test the consistency of the following statements.
    If Jane studies well, he will pass in exams.
    If Jane studies well, he will get a job.
    Succeeding in exam and getting a job simultaneously are not possible for him.
    Jane either enjoys or studies well.
    Finally, Jane enjoys.

11. If A works hard, then either B or C will enjoy themselves.
    If B enjoys himself, then A will not work hard.
    If D enjoys himself, then C will not.
    Therefore, if A works hard, then D will not enjoy himself.
    Show that the above statements constitute a valid argument by using the Rule CP.

12. Obtain the PCNF of the statement $S: (P \wedge Q) \vee (\neg P \wedge R)$. Using this, obtain the PCNF of $\neg S$ and hence the PDNF of S. Also determine the unique representation of the PCNF and PDNF.

13. Show that $R \wedge (P \wedge Q)$ is a valid conclusion from the premises.
$(P \vee Q), Q \rightarrow R, P \rightarrow M$, and $\neg M$.

14. Show that the following set of premises is inconsistent:
"If "Rama gets his degree, he will go for a job."
"If he goes for a job, he will get married soon."
"If he goes for higher study, he will not get married."
"Rama gets his degree and goes for higher study."

15. (*i*). Find the truth value of $(x)\ (P \rightarrow Q(x)) \vee (\exists x)\ R(x)$
where $P : 2 > 1$, $Q(x) : x > 3$, $R(x) : x > 4$, with the universe of discourse $E = \{2,3,4\}$.

(*ii*). Let the universe of discourse be $E = \{1,2,3,4,5,6\}$. The predicate formulae P(x) and Q(x) are defined by $P(x) : x \in \{2,3,6\}$ and $Q(x) : x<7$. Find the truth values of $P(4) \rightarrow (x)\ Q(x)$ and $(x)\ (P(x) \rightarrow Q(x))$.

16. (*a*). Write into the symbolic form using predicate symbols and quantifiers:
(*i*). There are some successful students who are not clever.
(*ii*). Every chalk board is black.

(*b*.) If S(x): x is a spy novel, L(x): x is long, and M(x): x is a mystery, then write the well-formed formula for the statements.
(*i*). All spy novels are long.
(*ii*). Not every mystery is a spy novel.
(*iii*). Some spy novels are not long.
(*iv*). Every long spy novel is a mystery.

17. Express the negation of the following statements using logical operators and quantifiers and also rewrite it in English.
(*i*). If the teacher is absent, then some students do not keep quiet.
(*ii*). All the students remains silent and the teacher is present.
(*iii*). Some students do not keep quiet or the teacher is absent.

18. Use the indirect method of proof to verify the following.
$(\forall x)\ (P(x) \vee Q(x)) \Rightarrow [((\forall x)\ P(x)) \vee ((\exists x)\ Q(x))]$

19. Show that $(\forall x)\ [(P(x) \vee Q(x)) \rightarrow R(x)] \Rightarrow (\forall x)[\ P(x) \rightarrow R(x)]$.

20. Verify the validity of the following arguments.
Every living thing is a plant or animal.
John's goldfish is alive and is not a plant.
All animals have hearts.
Therefore, John's goldfish has heart.

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Module 2
Algebraic Structures

1. Show that $\{Z^{+},*\}$ where * is defined by a*b=max(a,b), for all $a,b \in Z^{+}$, is a monoid. What is the identity element?

2. Prove that $(E, +)$ is a subgroup of the group $(Z, +)$, where Z is the set of integers and E is the set of even integers.

3. Check whether the set {5, 15, 25, 35} is a group or not with respect to multiplication module 40. If so,
   (i). form the Cayley's table,
   (ii). find the order of the group,
   (iii). find the identity element of the group,
   (iv). find the order of each element in a group,
   (v). find the inverse of each element in a group.

4. Let $S = \left\{\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}, \begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}, \begin{bmatrix}-1 & 0\\0 & -1\end{bmatrix}\right\}$. Show that S is a group under the operation of matrix multiplication.

5. Show that the set of all matrices of the form $\begin{bmatrix}x & x\\x & x\end{bmatrix}$, where x is nonzero real number is a group under the matrix multiplication.

6. The set S of all orderd pairs (a, b) of real numbers for which $a \neq 0$ w.r.t. the operation $\times$, defined by $(a, b) \times (c, d) = (ac, bc + d)$ is non-abelian. Let $H = \{(1, b) | b \in R\}$ is a subset of S. Show that H is a subgroup of $(S, \times)$.

7. Prove that a non-empty subset S of G is a subgroup of $(G,*)$ iff for any pair of elements $a, b \in S, a * b^{-1} \in S$.

8. Which of the following subsets of the group $M_{n\times n}$ are subgroups with respect the the matrix addition?
   (i). The set of all $n \times n$ singular matrices.
   (ii). The set of all $n \times n$ upper triangular matrices.

9. Show that the group $\{G, +_6\}$ is a cyclic group, where $G = \{0, 1, 2, 3, 4\}$ and find its generators.

10. Check whether the set {1, 3, 5, 7} under the binary operation $\times_6$ is cyclic group. List all properties of this algebraic structure.

11. (i). Show that every group of order 3 is cyclic.
    (ii). Prove that the group $G = \{1, -1, i, -i\}$ is cyclic under multiplication operator and find its generators.

12. If f is a homomorphism of a group G to a group G', then show that
(i). if H is a subgroup of G, then $f(H)$ is a subgroup of G'.
(ii). if H' is a subgroup of G', then $f(H') = \{x \in G | f(x) \in H'\}$ is a subgroup of G.

13. Prove that the following groups are isomorphic groups:
(i). $(\{1, -1\}, \times)$ and $(\{0,1\}, +_2)$
(ii). $(\{0,1,2,3\}, +_4)$ and $(\{1,2,3,4\}, \times_5)$

14. Find the left cosets of $\{[0], [2]\}$ in the group $(Z_4, +_4)$.

15. Let $G = \{1, b, b^2, b^3\}$ be a group and $H = \{1, b^2\}$ is a subgroup of G under multiplication. Find all the cosets of H, where $b^4 = 1$.

16. State and prove Lagrange's theorem on groups.

17. Show that $f: G \to G'$ defined by $f(x) = 2^x$ is a homomorphism, where $G = (R, +)$, $G' = (R^+, \cdot)$.

18. Find the code word generated by a parity check matrix
$H = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, when the encoding function is $e: B^3 \to B^6$.

19. For a (6, 3) code, the generated matrix G is shown below
$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$.
(i). Construct the parity check matrix.
(ii). Find all codewords generated by G.
(iii). Find the minimum Hamming distance.
(iv). Find the error if the received code is 1000111.

20. If $e: B^3 \to B^6$ be $(3, 6)$ encoding function, defined by $e(000) = 000000, e(001) = 001101, e(011) = 011011, e(010) = 011011, e(101) = 110101, e(100) = 101101, e(111) = 110010, e(110) = 110010.$
Check if this encoding function is a group code. List all the properties.

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Module 3
Counting Techniques

1. In a discrete mathematics class, every student is a major in computer science or mathematics, or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in this class?

2. Among a group of students, 49 study Physics, 37 study English and 21 study Biology. If 9 of these student's study Physics and English, 5 study English and Biology, 4 study Physics and Biology and 3 study Physics, English and Biology, find the number of students in the group.

3. A large software development company employs 100 computer programmers. Of them, 45 are proficient in Java, 30 in C#, 20 in Python, six in C# and Java, one in Java and Python, five in C# and Python, and just one programmer is proficient in all three languages above. Find the number of programmers those are not proficient in any of the three languages.

4. Suppose that there are 1807 freshmen at your school. Of these, 453 are taking a course in computer science, 567 are taking a course in mathematics, and 299 are taking courses in both computer science and mathematics. How many are not taking a course either in computer science or in mathematics?

5. A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken at least one of Spanish, French, and Russian, how many students have taken a course in all three languages?

6. In a box there are 10 black socks and 12 blue socks, and you need to get one pair of socks of the same colour. Supposing you can take socks out of the box only once and only without looking, what is the minimum number of socks you'd have to pull out at the same time in order to guarantee a pair of the same color?

7. Find the minimum number of students in a class such that three of them are born in the same month?

8. A basket of fruit is being arranged out of apples, bananas, and oranges. What is the smallest number of pieces of fruit that should be put in the basket in order to guarantee that either there are at least 8 apples or at least 6 bananas or at least 9 oranges?

9. Pick three natural numbers. Prove that there exist a pair whose sum or difference is divisible by 3.

10. During a month with 30 days, a baseball team plays at least one game a day, but no more than 45 games. Show that there must be a period of some number of consecutive days during which the team must play exactly 14 games.

11. Solve the following problems.
   (i). Assuming that repetitions are not permitted, how many four-digit numbers can be formed form the six digits 1, 2, 3, 5, 7, 8?
   (ii). How many of these numbers are less than 4000?
   (iii). How many of the numbers in part (a) are even?
   (iv). How many of the numbers in part (a) are odd?
   (v). How many of the numbers in part (a) are multiples of 5?
   (vi). How many of the numbers in part (a) contain both the digits 3 and 5?

12. There are 3 piles of identical red, blue and green balls, where each pile contains at least 10 balls. In how many ways can 10 balls be selected,
   (i). if there is no restriction?
   (ii). if at least one red ball must be selected?
   (iii). if at least one red ball, at least 2 blue balls and at least 3 green balls must be selected?
   (iv). if exactly one red ball must be selected?
   (v). if exactly one red ball and at least one blue ball must be selected?
   (vi). if at most one red ball is selected?
   (vii). if twice as many red balls as green balls must be selected?

13. Suppose 5 balls are to be placed in 3 boxes. Each can hold all the 5 balls. In how many different ways can we place the balls so that no box is left empty, if
   (*i*). balls and boxes are different?
   (*ii*). balls are identical, and boxes are different?
   (*iii*). balls are different, and boxes are identical?
   (*iv*). balls as well as boxes are identical?

14. Solve the following:
   (i). In how many ways can 6 boys and 4 girls sit in a row?
   (ii). In how many ways can they sit in a row if the boys are to sit together and the girls are to sit together?
   (iii). In how many ways can they sit in a row if the girls are to sit together?
   (iv). In how many ways can they sit in a row if just the girls are to sit together?

15. Solve the recurrence equation $c_n = 2c_{n-1} + 1; c_1 = 7$.

16. Solve the recurrence equation $c_n = -6c_{n-1} - 9c_{n-2}; c_1 = 2.5, c_2 = 4.7$.

17. Solve the recurrence equation $c_n = 2c_{n-1} + n^2; c_0 = 0$.

18. Solve the recurrence equation $c_n = 3c_{n-1} - 2c_{n-2} + 2^n; c_0 = 1, c_1 = 2$.

19. Solve the recurrence relation $c_n = \frac{1}{2}c_{n-1} + 2^{n-1}; c_0 = 0, n \geq 1$.

20. Using Generating Function Method, solve the recurrence relation $3c_n = 10c_{n-1} - 3c_{n-1} + 3^n; c_0 = 0, c_1 = 1, n \geq 2$.

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Module 4
Lattices and Boolean Algebra

1. Check whether the following relation is a partial order.
   (i). R is the relation on the set A = {1, 2, 4, 6, 8} defined by $a\ R\ b$ if and only if $\frac{b}{a}$ is an integer.
   (ii). For any real numbers a and b, $\underline{a\ R\ b}$ if and only if $a \leq b^3$.

2. For the POSET ⟨{3, 5, 9, 15, 24, 45}, divisor of⟩, find
   (i). the maximal and minimal elements.
   (ii). the greatest and the least elements.
   (iii). the upper bounds and LUB of {3, 5}.
   (iv). the lower bounds and GLB of {15, 45}.

3. Show that the set {2,3,6,12}, under the relation "multiple of" forms a lattice. Draw the Hasse diagram and find the greatest lower bound and least upper bound for the elements 6 and 12.

4. Determine whether the following POSETS are lattices.
   (i). {(1, 3, 6, 9, 12), D}
   (ii). {(1, 5, 25, 125), D},
   where D denotes the relation 'division'. Justify with the detailed answer.

5. Draw the Hasse diagram for the power set $P(S)$ of the set $S = \{a, b, c, d\}$, ordered by the subset relation, and identify the maximal, minimal, greatest, and least elements of the POSET. Also, find the greatest lower bound and least upper bound for the subsets $\{a\}$ and $\{b, c\}$.

6. Reduce the Boolean expression, $A = X{\cdot}Y + X{\cdot}(Y+Z) + Y{\cdot}(Y+Z)$ using Boolean algebra properties.

7. Prove the following Boolean identities.
   (i). $a{\cdot}(a' + b) = a{\cdot}b$.
   (ii). $a{\cdot}b + a{\cdot}b' = a$.

8. Simplify the following expressions using Boolean algebra.
   (i). $a'{\cdot}b{\cdot}(a'+ c) + a{\cdot}b'{\cdot}(b'+ c)$.
   (ii). $a + a'{\cdot}b{\cdot}c'+ (b + c)'$.

9. Find the disjunctive normal forms of the Boolean expression $f(x,y,z) = (x'{\cdot}y)'{\cdot}(x + y)$, using ($i$). the truth table method and (ii). the algebraic method.

10. Find the disjunctive normal forms of the Boolean expression $f(x,y,z)$ = = $x{\cdot}y + y{\cdot}z'$, using (i). the truth table method and (ii). the algebraic method.

11. Find the conjunctive normal forms of the Boolean expression $f(x,y,z)$ = = $x{\cdot}z + x'{\cdot}y + y{\cdot}z$, using ($i$). the truth table method and (ii). the algebraic method.

12. If R is a relation on the set {1, 2, 3, 4, 5}, list the ordered pairs in R, where
(*i*). $a\ R\ b$, if 3 divides $(a-b)$,
(*ii*). $a\ R\ b$, if $a^2 = b$.

13. Draw the digraph representing the partial ordering $\{(a,b) : a \text{ divides } b\}$ on the set {1, 2, 3, 4, 5, 6, 7, 8}. Reduce it to the Hasse diagram, representing the given partial ordering.

14. Prove $(a+b)' = a'\cdot b'$ and $(a\cdot b)' = a'+b'$, for all $a,\underline{b} \in$ B.

15. In any Boolean algebra, show that $\underline{a}\cdot b' + a'\cdot b = 0$ if and only if $a = b$.

16. Find the dual of the following Boolean expressions.
(a). $a'\cdot b\cdot c' + a'\cdot b'\cdot c$,
(b). $a\cdot(b'\cdot c' + b\cdot c)$.

17. If R is a relation from A = {1, 2, 3} to B = {1, 2} such that $a\ R\ b$ if $a > b$, construct the matrix representation of R.

18. The relation R on the set A = {1, 2, 3, 4, 5} is defined by the rule $(a,b) \in$ R, if 3 divides $(a-b)$.
(*i*). List the elements of R.
(*ii*). Find the domain and range of R.
(*iii*). List the elements of the complement of R.

19. Simplify the following Boolean function.
$f(a,b,c,d) = \Sigma(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11)$.

20. Simplify the following Boolean function.
$f(a,b,c) = a\cdot b\cdot c' + a\cdot b'\cdot c + a\cdot b'\cdot c' + a'\cdot b\cdot c + a'\cdot b'\cdot c$.

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